e) $x_n = \frac{n^2+24}{n+1}$

$$x_{n+1} - x_n = \frac{(n+1)^2+24}{n+2} - \frac{n^2+24}{n+1} =$$

$$= \frac{(n+1)^3+24(n+1) - (n^2+24)(n+2)}{(n+1)(n+2)} =$$

$$= \frac{n^3+3n^2+3n+1+24n+24-n^3-2n^2-24n-48}{(n+1)(n+2)} =$$

$$= \frac{n^3+3n-23}{(n+1)(n+2)}$$

$\uparrow$ $>0 \ \forall n \in \mathbb{N}$

$n^3+3n-23 \geq 0$

$D = 9+92 = 101$

$n_{1,2} = \frac{-3 \pm \sqrt{101}}{2}$

$n_1 = \frac{-3+\sqrt{101}}{2} \approx 3,5 > 0$

$n_2 = \frac{-3-\sqrt{101}}{2} \approx -6,5 < 0$

При $n \geq 4$ $x_n \uparrow$

g) $X_n = 3^n - 2^n$

$$X_{n+1} - X_n = (3^{n+1} - 2^{n+1}) - (3^n - 2^n) =$$

$$= 3 \cdot 3^n - 2 \cdot 2^n - 3^n + 2^n = 2 \cdot 3^n - 2^n =$$

$$= \underbrace{2^n}_{>0} \left(2 \cdot \frac{3^n}{2^n} - 1\right) \overset{?}{>} 0$$

при $n=1$: $2 \cdot \frac{3}{2} - 1 = 2$

при $n=2$: $2 \cdot \frac{9}{4} - 1 = 3{,}5$ $\Rightarrow \left(2 \cdot \left(\frac{3}{2}\right)^n - 1\right) > 0 \Rightarrow$

$$\Rightarrow 2^n \left(2 \cdot \left(\frac{3}{2}\right)^n - 1\right) > 0 \Rightarrow X_n \uparrow$$

i) $x_n = \frac{2^n}{n}$

$$x_{n+1} - x_n = \frac{2^{n+1}}{n+1} - \frac{2^n}{n} = \frac{2 \cdot 2^n \cdot n - 2^n(n+1)}{n^2+n} =$$

$$= \frac{2^1 \cdot 2^n \cdot n - 2^n n - 2^n}{n^2+n} = \frac{2^n(2n-n-1)}{n^2+n} =$$

$$= \frac{\overbrace{2^n}^{>0}\overbrace{(n-1)}^{\geq 0 \ \forall n \in \mathbb{N}}}{\underbrace{n(n+1)}_{>0 \ \forall n \in \mathbb{N}}} \Rightarrow x_n \uparrow$$

j) $x_n = \ln(n+1) - \ln n$

$$x_{n+1} - x_n = (\ln(n+2) - \ln(n+1)) - (\ln(n+1) - \ln n) =$$

$$= \ln \frac{n+2}{n+1} - \ln \frac{n+1}{n} = \ln \frac{n(n+2)}{(n+1)^2} \overset{?}{>} 0$$

$$\ln \frac{n(n+2)}{(n+1)^2} > \ln 1$$

M.P.

$$\underbrace{(e-1)}_{>0} \cdot \left(\frac{n(n+2)}{(n+1)^2} - 1\right) > 0$$

$$\frac{n(n+2) - (n+1)^2}{(n+1)^2} > 0$$

$$\frac{n^2+2n-n^2-2n-1}{(n+1)^2} > 0$$

$$\frac{-1}{(n+1)^2} > 0 \Rightarrow x_n \downarrow$$

a) $\sup(X+Y) = \sup X + \sup Y$

1) $\forall \begin{matrix} x \in X \\ y \in Y \end{matrix} : x + y \leq \sup X + \sup Y$

2) $\exists \varepsilon > 0 \ \exists x \in X : x > \sup X - \frac{\varepsilon}{2}$

$\exists y \in Y : y > \sup Y - \frac{\varepsilon}{2}$

$x + y > \sup X + \sup Y - \varepsilon$

$\sup(X)$:

1) $\forall x \in X : x \leq \sup(X)$

2) $\forall \varepsilon > 0 : \exists x \in X : x > \sup(X) - \varepsilon$

$\inf(X)$:

1) $\forall x \in X : x \geq \inf(X)$

2) $\forall \varepsilon > 0 : \exists x \in X : x < \inf(X) + \varepsilon$

1. $-X = \{-x \mid x \in X\}$

$\inf(X) \leq \forall x \in X \quad / \cdot (-1)$

$\cancel{-\inf(X) \geq -x : \overset{x \in X}{\Rightarrow} -\inf(X)}$

$-\inf(X) \geq \forall x \in -X$

2. $\forall \varepsilon > 0 : \exists x \in X : x < \inf(X) + \varepsilon \Rightarrow$

$\Rightarrow -x > -\inf(X) - \varepsilon \Rightarrow$

$\Rightarrow \forall \varepsilon > 0 : \exists x \in -X : -\inf(X) - \varepsilon < x$

$\left.\begin{array}{l} 1. \\ 2. \end{array}\right\} \Rightarrow -\inf(X) = \sup(-X)$ по опр.

d) $x_n = \frac{1-n}{\sqrt{n^2+1}}$

$$\frac{x_{n+1}}{x_n} = \frac{1-(n+1)}{\sqrt{(n+1)^2+1}} \cdot \frac{\sqrt{n^2+1}}{1-n} = \left\{ \frac{-n \cdot \sqrt{n^2+1}}{\left(\sqrt{(n+1)^2+1}\right)(1-n)} \right\}^2$$

$$\cancel{= \frac{-n \cdot \sqrt{n^2+1}}{}}$$

$$\left(\frac{-n \cdot \sqrt{n^2+1}}{\sqrt{(n+1)^2+1}\,(1-n)}\right)^2 = \frac{n^2(n^2+1)}{((n+1)^2+1)(1-n)^2} =$$

$$= \frac{n^4+n^2}{(n+1)^2(1-n)^2+(1-n)^2} = \frac{n^4+n^2}{n^4-n^2-2n+2} > 1$$

$$\cancel{n^2 > -n^2-2n+2}$$

$$\cancel{n^2+n-1>0}$$

$$\cancel{\left(n+\frac{1}{2}\right)^2 > \frac{5}{4}}$$

$$\frac{n^4+n^2-(n^4-n^2-2n+2)}{n^4-n^2-2n+2} > 0$$

$$\frac{\cancel{n^4}+n^2-\cancel{n^4}+n^2+2n-2}{n^4-n^2-2n+2} > 0$$

$$\frac{2n^2+2n-2}{n^4-n^2-2n+2} > 0$$

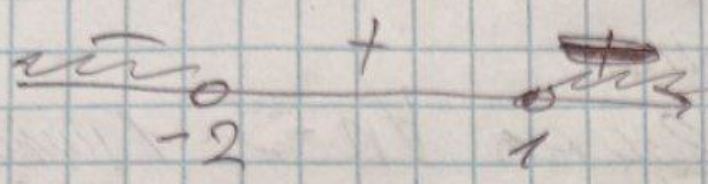
$$\frac{-2(-n^2-n+2)}{n^4-n^2-2n+2} > 0 \quad /:(-2)$$

$$\frac{\quad}{n^4-n^2-2n+2} < 0$$

1) $-n^2-n+2=0$ 2) $n^4-n^2-2n+2=0$

$n_1=-2$ $n=1$

$n_2=1$




$\Rightarrow X_n \downarrow$

Т.К. при $n \geq 2$

$x_n < 0$

$\sup(X_n) = 0$:

1) 1. При $n=1$: $X_1 = \frac{1-1}{\sqrt{1^2+1}} = 0$

2. При $n \geq 2$ $1-n<0 \ \forall n \geq 2$ $\Rightarrow$

$\sqrt{n^2+1} > 0 \ \forall n \geq 2$

$\Rightarrow \forall n, X_n \leq 0$, 0 - верхняя грань

2) $\exists \varepsilon > 0 : \exists x \in X_n : x > 0 - \varepsilon$

Т.К. $0 \in X_n$ ~~верхн~~ $\Rightarrow 0 > 0 - \varepsilon$

ч.т.д.

$\max(X_n) = 0$

$\inf(X_n) = -1$

1.

$$\frac{1-n}{\sqrt{n^2+1}} \geq -1$$

$$\sqrt{n^2+1} \geq n-1$$

$$n^2+1 \geq (n-1)^2$$

$$n^2+1 \geq n^2-2n+1$$

$2n \geq 0$ - верно $\forall n \Rightarrow -1$ нижняя грань

2. $\nexists \exists \varepsilon > 0 \ \nexists x \in X : x < -1+\varepsilon \Leftrightarrow$

$$\Leftrightarrow \frac{1-n+\sqrt{n^2+1}}{\sqrt{n^2+1}} < \varepsilon$$

Числ: $\dfrac{(\sqrt{n^2+1}-(n-1))(\sqrt{n^2+1}+(n-1))}{\sqrt{n^2+1}+(n-1)} =$

$$= \frac{(n^2+1)-(n-1)^2}{\sqrt{n^2+1}+(n-1)} =$$

$$= \frac{2n}{\sqrt{n^2+1}+n-1}$$

$$\frac{2n}{(\sqrt{n^2+1}+n-1)\sqrt{n^2+1}} < \varepsilon$$

Знам: $(\sqrt{n^2+1}+n-1)\sqrt{n^2+1}$

Т.К. $\sqrt{n^2+1} > \sqrt{n^2} = n \Rightarrow$

$$(\sqrt{n^2+1}+n-1)\sqrt{n^2+1} > (n+n-1)\cdot n = 2n^2-n$$

$$\frac{2n}{(\sqrt{n^2+1}+n-1)\sqrt{n^2+1}} < \frac{2n}{2n^2-n} = \frac{2}{2n-1} < \varepsilon$$

$$\frac{2}{2n-1} < \varepsilon$$

$$2 < \varepsilon(2n-1)$$

$$\frac{2}{\varepsilon} < 2n-1$$

$$n > \frac{1}{2}\left(\frac{2}{\varepsilon}+1\right)$$

такой $n$ существует $\Rightarrow$

$\Rightarrow -1 = \inf(X)$ ч.т.д.

Min(X):

~~Пусть $\exists x \in X_n$:~~

$\exists\, \exists n: x_n = -1$

$$\frac{1-n}{\sqrt{n^2+1}} = -1$$

$$1-n = -\sqrt{n^2+1}$$

$$n-1 = \sqrt{n^2+1}$$

$$n^2-2n+1 = n^2+1$$

$-2n = 0$

$n = 0 \Rightarrow$ противоречие $n \in \mathbb{N} \Rightarrow$

$\Rightarrow \min(X)$ не сущ.

$X_n$ - ограничено

е) $X_n = \frac{n^2 - n}{n+3}$

$$X_{n+1} - X_n = \frac{(n+1)^2 - (n+1)}{n+4} - \frac{n^2 - n}{n+3} =$$

$$= \cancel{n^2 + 2n + n - n - 1 - n^3}$$

$$= \frac{n^2 + 7n}{(n+4)(n+3)} > 0 \ \forall n \Rightarrow X_n \uparrow$$

$\inf(X_n) = \min(X_n) = 0$

$\nexists \max(X)$

$\sup(X_n) = +\infty$ $\Rightarrow$ неогр. сверху

д) $X_n = 5^n - 4^n$

$$X_{n+1} - X_n = (5^{n+1} - 4^{n+1}) - (5^n - 4^n) =$$

$$= 5 \cdot 5^n - 4 \cdot 4^n - 5^n + 4^n =$$

$$= 4 \cdot 5^n - 3 \cdot 4^n = \underbrace{4^n}_{>0}\left(4\left(\frac{5}{4}\right)^n - 3\right) \overset{?}{>} 0$$

$n=1$: $5 - 3 = 2$ $\Rightarrow$

$n=2$: $\frac{25}{4} - 3 = 3{,}25$

$\Rightarrow X_n \uparrow$

$\inf(X_n) = \min(X_n) = 1$

$\nexists \max(X)$

$\sup(X_n) = +\infty$

а) $X_n = \sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \dots \frac{1}{n(n+1)} =$

$= \frac{2-1}{1\cdot 2} + \frac{3-2}{2\cdot 3} + \dots + \frac{n+1-n}{n(n+1)} = 1 - \frac{1}{n+1}$

$\frac{1}{n+1}$ - убыв. $\Rightarrow$ $-\frac{1}{n+1}$ - возр.

$-\frac{1}{n+1} \approx 0$ при больших $n$ $\Rightarrow$

$\Rightarrow X_n \approx 1$ при больших $n$

$\sup(X_n) = 1$

$\triangleright$ 1. $X_n = 1 - \underbrace{\frac{1}{n+1}}_{>0} < 1 \Rightarrow 1$ - верхняя грань

2. $\forall \varepsilon > 0 : \exists x \in X_n : x > 1 - \varepsilon$

$1 - \frac{1}{n+1} > 1 - \varepsilon$

$n + 1 > \frac{1}{\varepsilon}$

$n > \frac{1}{\varepsilon} - 1$

$\blacktriangleleft \Rightarrow \sup(X_n) = 1$, $\max X_n \nexists$

$\inf(X_n) = \min(X_n) = \frac{1}{2}$

б) $X_n = \sum_{k=1}^{n} \frac{1}{4k^2-1} = \frac{1}{3} + \frac{1}{15} + \frac{1}{35} + \frac{1}{63} +$

$\frac{1}{4k^2-1} = \frac{1}{(2k-1)(2k+1)} = \frac{1}{2}\left(\frac{2}{(2k-1)(2k+1)}\right) =$

$= \frac{1}{2}\left(\frac{(2k+1)-(2k-1)}{(2k-1)(2k+1)}\right)$

$\cancel{X_n = \frac{1}{2}\sum_{k=1}^{n}\left(\frac{(2k+1)-(2k-1)}{(2k-1)(2k+1)}\right) = \frac{2}{3}}$

$X_n = \frac{1}{2}\sum_{k=1}^{n}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right) = 1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} \ldots$

$\ldots \frac{1}{2n+1} = \frac{1}{2}\left(1 - \frac{1}{2n+1}\right)$

$\sup(X_n) = \frac{1}{2}$

1. Т.к. $\frac{1}{2n+1} \downarrow \Rightarrow -\frac{1}{2n+1} \uparrow$

на больших $n$ $-\frac{1}{2n+1} \approx 0 \Rightarrow 1 - \frac{1}{2n+1} \approx 1 \Rightarrow$

$\Rightarrow \frac{1}{2}$ - верхняя грань

2. $\forall \varepsilon > 0 : \exists x \in X_n : x > \frac{1}{2} - \varepsilon$

$\frac{1}{2}\left(1 - \frac{1}{2n+1}\right) > \frac{1}{2} - \varepsilon \quad / \cdot 2$

$1 - \frac{1}{2n+1} > 1 - \varepsilon$

$2n+1 > \frac{1}{2\varepsilon}$

$n > \frac{1}{4\varepsilon} - \frac{1}{2} \Rightarrow \sup(X) = \frac{1}{2}$,

$\max(X_n) \nexists$

$\inf(X_n) = \min(X_n) = \frac{1}{3}$

d) $X_n = 1 - \left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots + \frac{1}{3^n}\right)$

$$S_n = \frac{b_1 - b_1 q^n}{1-q} = \frac{\frac{1}{3} - \frac{1}{3}\left(\frac{1}{3}\right)^n}{1 - \frac{1}{3}} =$$

$$= \frac{1}{3} \cdot \frac{1 - \left(\frac{1}{3}\right)^n}{\frac{2}{3}} = \frac{1}{2}\left(1 - \frac{1}{3^n}\right)$$

$$X_n = 1 - \frac{1}{2}\left(1 - \frac{1}{3^n}\right) = 1 - \frac{1}{2} + \frac{1}{2 \cdot 3^n} = \frac{1}{2} + \frac{1}{2 \cdot 3^n}$$

$\frac{1}{2 \cdot 3^n} \approx 0$ на больших $n \Rightarrow X_n \approx \frac{1}{2}$

$\sup(X) = \max(X_n) = \frac{1}{2} + \frac{1}{2 \cdot 3} = \frac{2}{3}$

$\inf(X_n) = \frac{1}{2}$

1. Нижняя грань — $\frac{1}{2}$ т.к.

2. $\forall \varepsilon > 0 \ \exists x \in X_n : x < \frac{1}{2} + \varepsilon$

$$\frac{1}{2} + \frac{1}{2 \cdot 3^n} < \frac{1}{2} + \varepsilon$$

$$1 < 2\varepsilon \cdot 3^n$$

$n > \log_3\left(\frac{1}{2\varepsilon}\right) \Rightarrow \inf(X_n) = \frac{1}{2}$, $\min(X_n) \nexists$

$A = \left\{ \frac{2+m}{3+n} \,/\, m, n \in \mathbb{N}, m \leq n \right\}$

$\sup(A)$

1. $2+m < 3+n \Rightarrow \frac{2+m}{3+n} < \frac{3+n}{3+n} = 1$ – верхняя граница

2. $\forall \varepsilon > 0 \ \exists m, n \in \mathbb{N}, m \leq n: \frac{2+m}{3+n} > 1 - \varepsilon$

$$\frac{2+m}{3+n} = \frac{(3+n)-(n-m+1)}{3+n} = 1 - \frac{n-m+1}{3+n}$$

$$1 - \frac{n-m+1}{3+n} > 1 - \varepsilon$$

$$\frac{n-m+1}{3+n} < \varepsilon$$

$\exists\, m = n \Rightarrow \frac{1}{3+n} < \varepsilon \Rightarrow n > \varepsilon^{-1} - 3 \Rightarrow$

$\Rightarrow \sup(A) = 1$, $\max(A) \nexists$

$\inf(A)$

1. Дробь минимальна, когда $2+m$ – минимальное, а $3+n$ – макс.

Пусть $m = 1$, а $n$ может быть сколь угодно большим, $\Rightarrow$

$\frac{2+1}{3+n}$ на больших числах $\approx 0$ – нижняя граница

2. $\forall \varepsilon > 0 \ \exists n \in \mathbb{N} : \frac{3}{3+n} < 0 + \varepsilon$

$$\frac{3}{3+n} < \varepsilon$$

$$n > \frac{3}{\varepsilon} - 3 \Rightarrow \inf(A) = 0, \ \min A \nexists$$

$$X_n = (-1)^n \frac{n-1}{n}$$

Если $n$ - четное:

1. $\sup(X_n)$

$(-1)^n = 1 \ \forall n$

$\frac{n-1}{n} = 1 - \frac{1}{n} \approx 1$ на больших $n$ $\Big\} \Rightarrow 1$ - верхняя грань

2. $\forall \varepsilon > 0 \ \exists x \in X_n : x > 1 - \varepsilon$

$$\underbrace{(-1)^n}_{=1} \frac{n-1}{n} > 1 - \varepsilon$$

$$1 - \frac{1}{n} > 1 - \varepsilon$$

$$\cancel{-\frac{1}{n} > 1 - \varepsilon}$$

$$\cancel{-1 > n - n\varepsilon}$$

$$\cancel{1 > n(1-\varepsilon)}$$

$$\cancel{\frac{1}{1-\varepsilon} > n}$$

$$\cancel{n < \frac{1}{1-\varepsilon}}$$

$$-\frac{1}{n} > -\varepsilon$$

$$n > \frac{1}{\varepsilon} \Rightarrow \sup(X_n) = 1, \ \max(X_n) \nexists$$

Если $n$ - нечетное:

$\inf(X_n)$

1. $(-1)^n = -1 \ \forall n$ $\Big\}$ $-1$ ниж грань

2. $\frac{n-1}{n} = 1 - \frac{1}{n} \approx$ ~~[illegible]~~ 0 на маленьких $n$

2. $\forall \varepsilon > 0 \ \exists x \in X_n : x < -1 + \varepsilon$

$$\underbrace{(-1)^n}_{=-1} \frac{n-1}{n} < -1 + \varepsilon$$

$$-1 + \frac{1}{n} < -1 + \varepsilon$$

$$\frac{1}{n} < \varepsilon$$

$$1 < n\varepsilon$$

$$n > \frac{1}{\varepsilon} \Rightarrow \inf(X_n) = -1$$

$$\min(X_n) \nexists$$

$X_n = n^{\cos \pi n}$

Заметим, что $\cos \pi n = (-1)^n \Rightarrow$

$\Rightarrow X_n = n^{(-1)^n}$

Если $n$ - четное; то $x_n = n$.

Т.к. $n \in \mathbb{N} \Rightarrow X_n$ не ограничено сверху $\Rightarrow \sup(X_n) = +\infty$, $\max(X_n) \nexists$

Если $n$ - натуральное, то $x_n = \frac{1}{n}$

1. при больших $n$ $x_n \approx 0$ - нижняя грань

2. $\forall \varepsilon > 0 \ \exists x \in X_n : x < 0 + \varepsilon$

$\frac{1}{n} < \varepsilon$

$n > \frac{1}{\varepsilon} \Rightarrow \inf(X_n) = 0, \min(X_n) \nexists$